A105 sample paper — worked solutions

every option explained, every calculation stepped out. Section C answers were computed and checked, since the paper prints none.

Section A — multiple choice

15 questions

Q1 Which one of the following determines the identity of an atom?

✓	**Atomic number**	The atomic number is the **number of protons**, and that is what makes an element the element it is. Change it and you have a different element entirely.
✗	Mass number	Mass number is protons **plus neutrons**. Carbon-12 and carbon-14 have different mass numbers and are both still carbon, so this cannot be what fixes identity.
✗	Number of neutrons	Neutrons vary freely between **isotopes** of the same element. They change the mass, not the identity.
✗	Number of valence electrons	Valence electrons set the **chemistry**, and different elements can share the same count — every Group 1 metal has one. They also change when the atom becomes an ion, and an ion is still the same element.

IN PLAIN TERMS Protons are the atom's name tag. Neutrons change its weight, electrons change its behaviour, but only protons change what it **IS**.

Q2 Which type of radiation emits a helium nucleus?

✓	**Alpha radiation**	An alpha particle **is** a helium nucleus: 2 protons and 2 neutrons, written ${}^{4}_{2}He$ or α. The mass number falls by 4 and the atomic number by 2.
✗	Beta radiation	A beta particle is a fast **electron** from the nucleus. Mass number is unchanged; atomic number rises by 1.
✗	Gamma radiation	Gamma is a high energy **electromagnetic wave**, not a particle at all. It carries no mass and no charge, so neither number changes.
✗	Positron emission	A positron is an **anti-electron**, positively charged. Atomic number falls by 1; still nothing like a helium nucleus.

IN PLAIN TERMS Alpha is the heavy one. If a question mentions helium, or a mass drop of 4, it is alpha.

Q3 Which one of the following statements based on the VSEPR model is TRUE?

✓	**Lone pair-lone pair repulsion is the strongest repulsive force.**	A lone pair is held by **one** nucleus only, so its cloud spreads wider and pushes harder. Two of them together repel most strongly of all. The order is **lp-lp > lp-bp > bp-bp**.
✗	Bond pair-bond pair repulsion is the strongest.	This is the **weakest** of the three. A bonding pair is pulled tight between two nuclei, so it takes up the least room.
✗	Lone pair-bond pair repulsion is the strongest.	It sits in the **middle** of the order, stronger than bp-bp but weaker than lp-lp.
✗	All repulsive forces are equal.	If they were equal, every 4-domain molecule would show exactly 109.5°. In reality ammonia is 107.5° and water is 104.5° — the lone pairs are clearly squeezing harder.

IN PLAIN TERMS A lone pair is only tied down at one end, so it takes up more room and shoves harder. More lone pairs, smaller angle.

Q4	Which one of the following types of intermolecular interaction is the strongest?
✓	**Ion-dipole interaction**	A **full** charge attracting a partial charge. A whole ionic charge beats any partial one, which is why ionic solids dissolve so readily in water.
✗	Temporary dipole interaction	London forces — the **weakest** of the four. They come from momentary, accidental electron imbalances.
✗	Permanent dipole interaction	Stronger than London, but only **partial** charges on both sides, so weaker than hydrogen bonding and much weaker than an ion.
✗	Hydrogen bonding	**The tempting answer, and it is genuinely strong** — but it is still a specially strong **dipole-dipole** attraction between partial charges. An ion brings a full charge, so ion-dipole wins.

IN PLAIN TERMS Rank by how much charge is involved: full charge beats partial charge. **ION-DIPOLE > HYDROGEN BONDING > PERMANENT DIPOLE > LONDON**.

Q5	Boyle's law states that for a fixed amount of gas at constant temperature, the pressure of a gas is ____.
✓	**inversely proportional to its volume**	`PV = constant` Squeeze the gas into half the space and the molecules hit the walls twice as often, so the pressure doubles.
✗	directly proportional to its volume	That is backwards. If it were true, a compressed gas would push **less**, and a bicycle pump would not work.
✗	directly proportional to the square of its volume	No square appears anywhere in the gas laws. Invented relationship.
✗	independent of its volume	If pressure did not depend on volume, nothing could ever be compressed or pressurised.

IN PLAIN TERMS Smaller box, harder push. P and V are a seesaw: one up, the other down.

Q6	What is the dilution factor required to dilute 0.01 M HCl solution to 0.001 M?
✓	**10**	`DF = 0.01 ÷ 0.001 = 10` The dilution factor is simply how many times weaker it has to end up. Ten times weaker means one part acid to nine parts water, making ten parts in total.
✗	20	Would take it to 0.0005 M, twice as dilute as asked.
✗	100	Would give 0.0001 M. This is the answer to "0.01 to 0.0001" — one decimal place too far.
✗	200	Would give 0.00005 M, far past the target.

IN PLAIN TERMS Divide the concentration you have by the one you want. That is the whole calculation.

Q7	What is the pH of a 0.010 M solution of NaOH?
✓	**12.00**	NaOH is a strong **base**, so it is a two-step calculation. `[OH⁻] = 0.010 → pOH = −log(0.010) = 2.00 → pH = 14 − 2.00 = 12.00`
✗	2.00	**The trap, and the most common wrong answer.** This is the **pOH**, not the pH. Taking −log of 0.010 and stopping there forgets that the 0.010 was a hydroxide concentration.
✗	7.00	That is neutral water. A 0.010 M alkali is nowhere near neutral.
✗	10.00	Alkaline, so at least the right side of 7, but arrived at by an arithmetic slip — there is no route from 0.010 M to exactly 10.00.

IN PLAIN TERMS For an alkali you always get pOH first, then subtract from 14. Stopping at the first number is the classic mistake.

Q8 Which one of the following glassware is appropriate for preparing an accurate standard solution?

✓	**Volumetric flask**	Calibrated to **contain** one precise volume at a stated temperature, marked by a single etched line. That is exactly what a standard solution needs.
✗	Measuring cylinder	Ordinary glassware with coarse graduations. Fine for approximate volumes, not for a solution whose concentration must be exact.
✗	Conical flask	Its markings are rough guides only. Its real job is to be **swirled** during a titration without splashing.
✗	Beaker	The least precise of the four. Its graduations are decorative more than quantitative.

IN PLAIN TERMS Volumetric apparatus — volumetric flask, bulb pipette, burette — is precise. Everything else is approximate.

Q9 What will happen to a reversible reaction when a catalyst is added?

✓	**Both forward and reverse reaction rates increase equally.**	A catalyst lowers the activation energy of the **same barrier** that both directions must cross, so both speed up by the same factor. Equilibrium is reached sooner, but in the same place.
✗	Only the forward reaction rate increases.	There is only one hill between reactants and products. Lowering it necessarily helps both directions — it cannot be lowered from one side only.
✗	Only the reverse reaction rate increases.	Same objection, mirrored.
✗	The equilibrium constant changes.	**Only temperature changes K.** A catalyst changes how fast equilibrium arrives, never where it sits, so the yield is unchanged.

IN PLAIN TERMS A catalyst is a shortcut through the hill that both directions can use. It changes the **SPEED**, never the destination.

Q10 What is the main characteristic of a buffer solution?

✓	**It resists changes in pH when small amounts of acid or base is added.**	A buffer holds a weak acid and its conjugate base together. The base half mops up added H^+ and the acid half mops up added OH^-, so the pH barely moves.
✗	It neutralizes strong acids completely.	"Completely" is too strong. A buffer **moderates** the change and has a finite capacity; enough acid will overwhelm it.
✗	It changes pH rapidly when water is added.	The opposite is true. Diluting a buffer changes both partners by the same factor, so their **ratio** is unchanged and the pH is nearly unmoved.
✗	It only works with strong acids and bases.	Backwards. A buffer must be built from a **weak** acid or base with its conjugate; a strong acid and its salt is not a buffer at all.

IN PLAIN TERMS A buffer is a chemical shock absorber. Two partners, one to catch acid and one to catch alkali.

Q11 What does a positive value of enthalpy change ($\Delta H > 0$) indicate?

✓	**Heat is absorbed by the chemical system during the reaction.**	Positive means the system **gained** energy, so it took heat in from the surroundings. That is endothermic, and the surroundings feel cold.
✗	Heat is released to the surroundings during the reaction.	That is exothermic, which is **negative** ΔH. The sign is the wrong way round.
✗	The reaction is at equilibrium.	Equilibrium is about ΔG and the rates matching. ΔH says nothing about whether a reaction has settled.
✗	The reaction occurs at low temperature.	ΔH is not a statement about temperature. An endothermic reaction can be run hot or cold.

IN PLAIN TERMS Think of the system's own account. Positive means energy paid **IN**; negative means energy paid **OUT**.

Q12 Which one of the following statements is TRUE?

✓	**The gaseous state has the highest entropy.**	Entropy counts how many ways the particles can be arranged. Gas particles are far apart and moving freely, which is by far the most disordered arrangement.
✗	The solid state has the highest entropy.	A solid is the **lowest**. Particles are locked in a lattice and can only vibrate about fixed points.
✗	The liquid state has the highest entropy.	A liquid sits in the middle: particles can slide past one another but are still touching.
✗	All three states have the same entropy.	If that were so, melting and boiling would involve no entropy change, and ΔS would never appear in a phase-change calculation.

IN PLAIN TERMS Order of freedom, always: **SOLID < LIQUID < AQUEOUS < GAS**.

Q13 According to the Collision Theory, which two conditions must be met in order for a successful chemical reaction to occur?

✓	**Correct orientation and sufficient kinetic energy**	The molecules must hit each other **hard enough** to clear the activation energy, and hit at the **right angle** for the bonds to rearrange. Both, or the collision simply bounces.
✗	High temperature and low pressure	These are conditions that **change** a rate, not the requirements for a single collision to succeed. Low pressure would slow a gas reaction anyway.
✗	Low activation energy and high molecular mass	Low activation energy helps, but it is a property of the reaction, not a condition on a collision. Molecular mass is irrelevant.
✗	Presence of a catalyst and high surface area	Both speed reactions up, but neither is **required** — countless reactions proceed with no catalyst at all.

IN PLAIN TERMS Hard enough, and square on. A glancing blow or a gentle nudge does nothing.

Q14 A redox reaction with a positive $E°_{cell}$ value indicates _____.

✓	**spontaneous reaction**	Positive $E°_{cell}$ and negative ΔG go together $\Delta G = -nFE$ so a positive cell potential means ΔG is negative and the reaction goes by itself.
✗	a non-spontaneous reaction	That is a **negative** $E°_{cell}$, which needs a power supply to drive it — an electrolytic cell.
✗	equilibrium has been reached	At equilibrium $E°_{cell}$ is **zero**, not positive. A flat battery.
✗	a very slow reaction	**Potential says nothing about speed.** Thermodynamics tells you whether it can happen; kinetics tells you how fast.

IN PLAIN TERMS Positive volts means the reaction will push electrons around a circuit on its own. That is what spontaneous means here.

Q15 A voltaic cell converts ___ energy to ___ energy through ___ redox reactions.

✓	**chemical, electrical, spontaneous**	A voltaic (galvanic) cell is a **battery**. A reaction that wants to happen anyway is used to push electrons through a wire, turning chemical energy into electrical.
✗	chemical, electrical, non-spontaneous	Right direction of energy, wrong driver. A non-spontaneous reaction cannot generate anything — it has to be forced.
✗	electrical, chemical, spontaneous	This is an **electrolytic** cell's energy direction, and even then the reaction is non-spontaneous, so both halves are wrong.
✗	electrical, chemical, non-spontaneous	A correct description — of an **electrolytic** cell, which is the other kind. The question asked about voltaic.

IN PLAIN TERMS Voltaic makes electricity from a reaction that wants to happen. Electrolytic uses electricity to force one that does not.

Section B — fill in the blanks

2 questions

Q16 Figure 1 shows the phase diagram of iodine. Fill in the blanks with the most appropriate answer.

a) There are ___ different phases in the phase diagram of iodine.

1 The diagram is divided into **three** labelled regions. — A phase diagram is a map: each region is one physical state.

2 Those regions are **solid**, **liquid** and **gas**. — Iodine has no exotic extra phases here, so it is just the three ordinary states.

3

b) The boiling point of iodine at 1.00 atm is ___ °C.

1 Draw a horizontal line across at **P = 1.00 atm**. — Boiling point is always quoted at a stated pressure, so fix the pressure first.

2 Find where that line crosses the **liquid-gas** boundary. — Boiling is the liquid to gas change, so it is that curve you want, not the solid-liquid one.

3 Read straight down to the temperature axis. — **113.7 is the melting point at 1.00 atm and 113.5 is the triple point — both are on the diagram as decoys.**

184 °C

c) The triple point of iodine is at ___ atm and ___ °C.

1 Find the single point where **all three** curves meet. — The triple point is the one condition where solid, liquid and gas coexist in equilibrium.

2 Read across for pressure and down for temperature. — Both coordinates are dashed onto the axes on the figure.

0.118 atm and 113.5 °C

d) The melting point of iodine ___ with increasing pressure.

1 Look at the **solid-liquid** line and ask which way it leans. — That line **is** the melting point, plotted against pressure.

2 It slopes up and to the **right**, so higher pressure sits at higher temperature. — This is the normal case for almost every substance. **Water is the famous exception — its line leans left — so do not answer from memory of water.**

increases

Q17 The rate law of a reaction between compounds X and Y is

```
Rate = k[X][Y]²
```

a) The overall order of the reaction is ___.

1 Read the exponents: X is to the power **1**, Y is to the power **2**. — An unwritten exponent is 1.

2

```
overall order = 1 + 2 = 3
```

— Overall order is just the sum of the individual orders.

3

b) If [X] increases by 4 times and [Y] is unchanged, the rate increases by a factor of ___.

1 X appears to the power 1, so the rate scales as — First order means the rate follows the concentration exactly.

```
4¹ = 4
```

4

c) If [Y] increases by 3 times and [X] is unchanged, the rate increases by a factor of ___.

1 Y appears to the power 2, so the rate scales as — Second order means the factor gets squared — this is where marks are usually lost.

```
3² = 9
```

9

d) The half-life of a reaction is the time required for the reactant's concentration to decrease to __ of the initial concentration.

1	**Half**, by definition.	The word itself says it. This mark is free.

half

e) The order with respect to Y is 2. Hence the half-life of Y is __ proportional to its initial concentration.

1	For second order, $t_{½} = 1 / (k[Y]_0)$	The starting concentration sits in the **denominator**.
2	So a larger starting concentration gives a **shorter** half-life.	That is what inversely proportional means. Compare first order, where t½ does not depend on concentration at all.

inversely

Section C — structured questions

5 questions

Q18 Sulfuryl chloride decomposes according to

$$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$$

a) Name the product in the backward reaction. 1 mark

1	The backward reaction runs **right to left**, so its product is what sits on the left.	Its reactants are SO_2 and Cl_2; the thing they re-form is the answer.

Sulfuryl chloride, SO_2Cl_2

b) Write the equilibrium constant expression. 1 mark

1	Products on top, reactants underneath, each raised to its coefficient.	Every coefficient here is 1, so no powers appear.
2	$K_c = [SO_2][Cl_2] \div [SO_2Cl_2]$	Concentrations **multiply**, never add.

$K_c = [SO_2][Cl_2] \div [SO_2Cl_2]$

c) i. State the direction of shift and how the amount of SO_2 changes when the pressure of the system increases. 2 marks

1	Count moles of **gas** on each side: left has **1**, right has **2**.	Only gases count for a pressure argument.
2	Raising the pressure shifts the equilibrium toward the side with **fewer** moles of gas, which is the **left**.	The system relieves the extra pressure by taking up less volume.
3	SO_2 is on the right, so it is being consumed.	Shifting left means the right-hand species decrease.

Shifts LEFT; the amount of SO_2 DECREASES

c) ii. State the direction of shift and how the amount of SO_2 changes when the concentration of SO_2Cl_2 increases. 2 marks

1	More reactant has been added on the **left**.	
2	The system opposes the change by consuming it, so the equilibrium shifts **right**.	Add to one side and the balance tips away from that side.
3	SO_2 sits on the right, so more of it is made.	

Shifts RIGHT; the amount of SO_2 INCREASES

Q19 A bicycle tyre, and the ideal gas law.

a) State the ideal gas law. 1 mark

1	$PV = nRT$	P pressure, V volume, n moles, R the gas constant, T temperature in **kelvin**.

PV = nRT

b) The tyre is inflated at 35 °C. Overnight the temperature drops to 15 °C and the tyre feels slightly flat. Explain, assuming the volume does not change. 2 marks

	Step	Comment
1	Identify what is fixed and what changes: **V** and **n** are constant, **T** falls.	The air is sealed in, so no gas escapes; the question fixes the volume.
2	Rearrange to isolate the two that vary: `P = (nR / V) × T`	With n, R and V all constant, the bracket is just a number, so **P is directly proportional to T**.
3	T falls from 308 K to 288 K, so **P falls**.	Physically: colder molecules move more slowly, so they strike the tyre wall less often and less hard.

The pressure drops because temperature dropped at constant volume, so the tyre feels flat

c) The cyclist sits on the bicycle and the tyre is compressed slightly. Explain, assuming the surrounding temperature is constant. 2 marks

	Step	Comment
1	Now **T** and **n** are constant and the applied **pressure** rises.	The rider's weight presses down on the tyre.
2	With nRT constant, `PV = constant` , which is Boyle's law.	P and V are inversely proportional.
3	P up therefore means **V down**: the tyre is squashed into a smaller volume.	The gas is compressed until its internal pressure balances the load.

Increased pressure at constant temperature reduces the volume, so the tyre compresses

Q20 TNT combustion.

$$4\ C_7H_5N_3O_6(s) + 21\ O_2(g) \rightarrow 28\ CO_2(g) + 10\ H_2O(g) + 6\ N_2(g)$$

Δ_fH: TNT −67, CO_2 −393.51, $H_2O(g)$ −241.82 kJ/mol. A_r: C 12.01, H 1.01, N 14.01, O 16.00

a) How many moles of TNT undergo combustion as shown in equation 1? 1 mark

	Step	Comment
1	Read the coefficient in front of the TNT.	The equation as written already tells you: **4**. No calculation needed — this mark is free.

4 mol

b) Calculate the enthalpy change of reaction for equation 1. 2 marks

	Step	Comment
1	Use $\Delta H = \Sigma n\Delta_fH(\text{products}) - \Sigma n\Delta_fH(\text{reactants})$	n is the coefficient from the balanced equation, so every term gets multiplied by it.
2	Products: $28(-393.51) + 10(-241.82) + 6(0)$ $= -11018.28 - 2418.20 + 0 = -13436.48$	**N_2 is an element in its standard state, so its Δ_fH is ZERO** — it contributes nothing.
3	Reactants: $4(-67) + 21(0) = -268$	O_2 is also an element, so also zero.
4	$\Delta H = -13436.48 - (-268) = -13436.48 + 268$	**Subtracting a negative adds.** This is where the sign is most often lost.

$\Delta_{rxn}H$ = −13168.48 kJ (exothermic, as any combustion must be)

c) Using your answers to (a) and (b), calculate the enthalpy change when 5 g of TNT is burnt. Answer to 2 decimal places. 5 marks

1 Molar mass of $C_7H_5N_3O_6$:

```
7(12.01) + 5(1.01) + 3(14.01) + 6(16.00)
= 84.07 + 5.05 + 42.03 + 96.00 = 227.15 g/mol
```

Count every atom in the formula, including all six oxygens.

2 Moles in 5 g:

```
n = m ÷ M = 5 ÷ 227.15 = 0.0220119 mol
```

Keep the full precision here; round only at the very end.

3 **The step that carries the marks:** the −13168.48 kJ from part (b) is for **4 mol**, not one.

```
per mole = -13168.48 ÷ 4 = -3292.12 kJ/mol
```

This is why part (a) was asked at all — it sets up this division. Skipping it makes the answer four times too big.

4

```
ΔH = -3292.12 × 0.0220119 = -72.4658 kJ
```

Energy released scales directly with how much fuel is burnt.

5 Round to 2 decimal places as instructed.

The question says so explicitly, and the instruction carries a mark.

$\Delta H = -72.47$ kJ

Q21 Aluminium is extracted from molten Al_2O_3 in an electrolytic cell. Electrode A:

```
Al³⁺ + 3e⁻ → Al
```

Electrode B:

```
2O²⁻ → O₂ + 4e⁻
```

A current of 15 000 A flows for 2 hours. F = 96 500 C/mol, M(Al) = 26.98 g/mol.

a) State which carbon electrode (A or B) is the anode and which is the cathode. 2 marks

1 Electrode A **gains** electrons ($3e^-$ on the left), so it is **reduction**.

Reduction Is Gain — the RIG half of OIL RIG.

2 Reduction always happens at the **cathode**.

This is true in BOTH cell types. Only the +/− signs swap between galvanic and electrolytic.

3 Electrode B **loses** electrons ($4e^-$ on the right), so it is oxidation, at the **anode**.

Consistent with the stem: aluminium forms at the cathode and oxygen at the anode.

A = CATHODE (reduction) B = ANODE (oxidation)

b) Write the overall chemical equation for the cell. 2 marks

1 The two halves must exchange the **same number** of electrons. A uses 3, B releases 4, and the lowest common multiple is **12**.

Electrons cannot be created or destroyed in the overall equation.

2 Multiply A by 4:

```
4Al³⁺ + 12e⁻ → 4Al
```

3 Multiply B by 3:

```
6O²⁻ → 3O₂ + 12e⁻
```

4 Add them and cancel the 12 electrons from both sides.

The electrons must not appear in the final equation.

$4Al^{3+} + 6O^{2-} \rightarrow 4Al + 3O_2$ (equivalently $2Al_2O_3 \rightarrow 4Al + 3O_2$)

c) Calculate the mass of aluminium (in kg) produced at the cathode, to two decimal places. 4 marks

1 Convert the time to **seconds**:

```
2 h × 3600 = 7200 s
```

Q = It requires seconds. Leaving it in hours is the single most common error in this question.

2 Charge passed:

```
Q = I × t = 15000 × 7200 = 1.08 × 10⁸ C
```

Current is coulombs per second, so amps times seconds gives coulombs.

3 Moles of electrons:

```
n(e⁻) = Q ÷ F = 1.08 × 10⁸ ÷ 96500 = 1119.171 mol
```

One mole of electrons carries 96 500 C — that is what the Faraday constant means.

4 **Divide by the electrons in the half-equation:** Al^{3+} needs **3** electrons per atom.

```
n(Al) = 1119.171 ÷ 3 = 373.057 mol
```

This is the step that separates a 4-mark answer from a 2-mark one. Silver would divide by 1, aluminium by 3.

5 Convert to mass:

```
m = n × M = 373.057 × 26.98 = 10065.08 g
```

6 Convert to kilograms as asked:

```
10065.08 ÷ 1000 = 10.065 kg
```

The question specifies kg **and** two decimal places.

m(Al) = 10.07 kg

Q22 For

```
XY₂ ⇌ X + Y₂
```

, Figure 3 plots log(rate) against log[XY_2] with points (0.0000, 1.0000), (0.3010, 1.6021), (0.6990, 2.3979), (1.0000, 3.0000) and (1.3010, 3.6021).

a) With reference to Figure 3, determine the rate law for the forward reaction. 3 marks

1 Take logs of a rate law and it becomes a straight line:

```
rate = k[XY₂]ⁿ
log(rate) = n × log[XY₂] + log k
```

That is **y = mx + c**, so the **gradient is the order** and the **intercept is log k**. This is the whole point of plotting logs.

2 Gradient from the two end points:

```
(3.6021 − 1.0000) ÷ (1.3010 − 0.0000)
= 2.6021 ÷ 1.3010 = 2.00
```

Use the widest pair of points available — it minimises reading error.

3 Intercept is where log[XY_2] = 0, read straight off the first point:

```
log k = 1.0000, so k = 10¹ = 10
```

log[XY_2] = 0 means [XY_2] = 1, so the rate there **is** k.

4 Check it against another point: at log[XY_2] = 0.3010, [XY_2] = 2, so

```
rate = 10 × 2² = 40
```

and

```
log 40 = 1.6021
```

That matches the plotted point exactly, so the rate law is right.

rate = 10[XY_2]^2 (second order in XY_2, k = 10 $M^{-1}s^{-1}$)

b) Given that the backward reaction is first order with respect to both X and Y_2, form an equation relating the forward and backward rates at equilibrium. 1 mark

1 At equilibrium the two rates are **equal**.

That is the definition of dynamic equilibrium: both reactions still run, at matched speeds, so nothing appears to change.

2 Forward

```
= k_f[XY₂]²
```

and backward

```
= k_b[X][Y₂]
```

First order in each of X and Y_2 means each appears to the power 1.

$k_f[XY_2]^2 = k_b[X][Y_2]$

c) At equilibrium [X] = 0.288 M, $[Y_2]$ = 0.288 M and $[XY_2]$ = 0.356 M. Given A = 1.5 × 10^{17} s^{-1}, T = 500 K and R = 8.314 × 10^{-3} kJ $mol^{-1}K^{-1}$, calculate the activation energy of the backward reaction, to the nearest whole number. 4 marks

1 Rearrange the equation from (b) to isolate the backward rate constant:

$$k_b = k_f[XY_2]^2 \div ([X][Y_2])$$

Part (b) exists to give you this. Everything on the right is now known.

2 Substitute, using k_f = 10 from part (a):

$$k_b = 10 \times (0.356)^2 \div (0.288 \times 0.288)$$

$$= 10 \times 0.126736 \div 0.082944 = 15.280$$

The three parts of this question are a chain — (a) feeds (b), and (b) feeds (c).

3 Now Arrhenius:

$$k = Ae^{-Ea/RT}$$

, rearranged to

$$E_a = -RT \ln(k \div A)$$

Take natural logs of both sides and solve for E_a.

4

$$k \div A = 15.280 \div 1.5 \times 10^{17} = 1.0187 \times 10^{-16}$$

$$\ln(1.0187 \times 10^{-16}) = -36.823$$

5

$$RT = 8.314 \times 10^{-3} \times 500 = 4.157 \text{ kJ/mol}$$

$$E_a = -4.157 \times (-36.823) = 153.07 \text{ kJ/mol}$$

R was given in kJ, so the answer is already in kJ/mol — no conversion. The two minus signs multiply to a plus, as they must, since activation energy is always positive.

E_a = 153 kJ/mol